

CS305: Computer Networking

2025 Fall Semester Written Assignment # 1

Please answer questions in English. Using any other language will lead to a zero point.

Q 1. List the five layers of the Internet protocol stack. For each layer, offer a concise explanation using one sentence, and give an example of a protocol or specific networking technology of that layer.

Solution:

- Application Layer: user-to-user exchange of messages (e.g., SMTP) (network applications and services).
- Transport Layer: end-to-end exchange of segments (e.g., TCP) (process-to-process, using sockets).
- Network Layer: host-to-host movement of packets (e.g., IPv4) (datagrams).
- Datalink Layer: hop-by-hop exchange of frames (e.g., WiFi).
- Physical Layer: raw transmission of bits (e.g., optical fiber).

Q 2. For the following pairs of technical terms, provide a definition for each and explain the primary differences between them. Your response should be clear and concise.

- (a) “Circuit-switching” and “packet-switching”
- (b) “Client-server” and “peer-to-peer”

Solution:

- (a) Circuit-switching sets up a dedicated communication path between two devices for the entire duration of a session, as in traditional telephone calls. Packet-switching, used in the Internet, breaks data into packets that share network paths and are reassembled at the destination. The main difference is that circuit-switching reserves a fixed path and bandwidth, while packet-switching dynamically shares resources among many users.
- (b) Client-server networks have a central always-on server that provides resources or services to multiple client devices that request them. In peer-to-peer networks, there is no always-on server. Each peer connects to others, sharing resources directly. The main difference is that client-server systems are centralized, while peer-to-peer systems are decentralized and distribute control among all nodes.

Q 3. Answer the following questions:

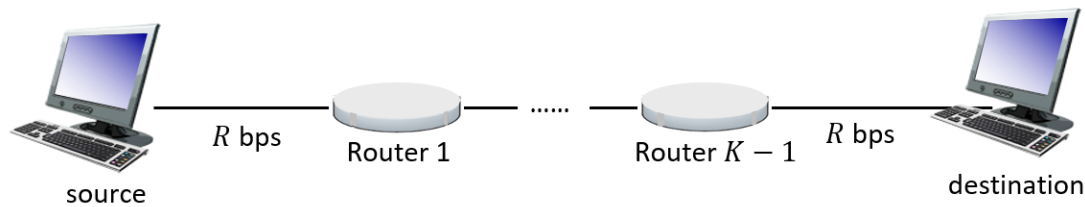
- (a) What type of applications is TCP better suited for? Any application examples?
- (b) What type of applications is UDP better suited for? Any application examples?

Solution:

- (a) Those applications that require reliable data transfer, e.g., email, file transfer.
- (b) Those applications that are time-sensitive, e.g., real-time video streaming, real-time gaming.

Students may have other answers. He or she can get the points as long as the answer is reasonable.

Q 4. Consider a packet of L bits sending from the source to destination through a K -hop path. That is, there are $K - 1$ routers between the source and destination. Suppose each link has a transmission rate of R bits per second (bps), and the propagation delay is d for each hop.



- Consider a packet switching network. Suppose there is no nodal processing delay and queuing delay. What is the end-to-end delay?
- Consider a circuit switching network. Suppose the circuit setup time is τ seconds. Links in the network use frequency division multiplexing (FDM), where the associated frequency band is divided into F subbands, each being allocated to a user (or circuit). [Hint: As a result, the transmission rate allocated to each user (or circuit) is R/F .] What is the end-to-end delay?
- Consider a packet switching network with $L = 1000$ bits, $K = 2$, $R = 20$ Mbps, $d = 10\mu s$. There are two packets sent one after the other, and there are no other packet in the system. Let the nodal processing delay at the router be $5\mu s$. We ignore the nodal processing delay at the source and destination. Compute the time required to send both packets from the source and destination. [Note: In our lecture, we set 1 Kbit = 10^3 bits and 1 Mbit = 10^6 bits.]

Solution:

- The one-hop delay is equal to the one-hop transmission delay plus the one-hop propagation delay, i.e., $L/R + d$. Since packet switching exploits a store-and-forward scheme, the end-to-end delay along this K -hop path is equal to $K(L/R + d)$.
- Since the links uses FDM with F subbands, the transmission rate of each subband is R/F bps. It takes τ seconds for circuit setup, takes $L/(R/F)$ seconds to transmit the packet, and Kd seconds for each bit to propagate. Thus, the end-to-end delay is equal to $\tau + L/(R/F) + Kd$.
- Consider the first packet. It is received by the router at $L/R + d = 50 + 10 = 60\mu s$. The router starts to send the first packet at $60 + 5 = 65\mu s$ and finishes the sending at $65 + L/R = 115\mu s$. The destination completely receives the first packet at $115 + d = 125\mu s$. Now, consider the second packet. It is received by the router at $2L/R + d = 110\mu s$. Then, it is sent by the router at $\max\{115, 110 + 5\}\mu s = 115\mu s$. The second packet receives by the destination at $115 + L/R + d = 175\mu s$.

Q 5. Consider a set of packets with size 10 Mbits and a queue. These packets arrive at the queue with certain patterns defined in (a) and (b), waiting for transmission. The transmission rate is 10 Mbps.



- Suppose there is one packet arrival every second. What is the average queuing delay of these packets?
- Suppose K packets arrive simultaneously every K seconds. What is the average queuing delay of these packets?
- What are the traffic intensity of the scenarios considered in (a) and (b)? Any insights?

Solution:

- Without loss of generality, suppose the first packet arrives at time $t = 0$ second. Since the queue is empty, it does not need to wait. The first packet is completely sent at time $t = 1$ second. Then, according to the description, the second packet arrives at time $t = 1$ second. Since the first packet has been sent completely, the second packet does not need to wait. The second packet is completely sent at time $t = 2$ second, so and so forth. As a result, when a new packet arrives, all previous packets have been sent, so that packet can be sent immediately without queuing delay. That is, the average queuing delay is zero.

- (b) Without loss of generality, suppose the first K packets arrive at time $t = 0$ second simultaneously. Then, the first packet in the queue is sent immediately after its arrival. The second packet waits for 1 seconds in the queue. The k^{th} packet waits for $(k - 1)$ seconds. Thus, the average queuing delay for these K packets arrived at time slot 0 is equal to

$$\frac{1}{K} \sum_{k=1}^K (k - 1) = \frac{(K - 1)}{2} \quad (1)$$

Then, at time K seconds, all the aforementioned K packets have been sent completely, and K new packets arrive at the queue simultaneously. It is easy to show that these K new packets have an average queuing delay of $(K - 1)/2$, so and so forth. Thus, the average delay is $(K - 1)/2$.

- (c) In (a), the traffic intensity is $(1 \times 10Mbps)/10Mbps = 1$. In (b), the traffic intensity is

$$(K \times 10Mbps/K)/10Mbps = 1.$$

Although two scenarios have the same traffic intensity, the average queuing delay can be different. The average queuing delay highly depends on the arrival patterns of the packets. Under the bursty traffic in (b), the average queuing delay tends to be larger. The queuing delay can vary from packet to packet.

Q 6. Consider the following message and answer questions.

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host:
gaia.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (Windows;U; Windows NT
5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax)
<cr><lf>Accept:ext/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5<cr><lf>Accept-Language:
en-us, en;q=0.5<cr><lf>Accept-Encoding: zip,deflate<cr><lf>Accept-
Charset: ISO-8859-1, utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive:
300<cr><lf>Connection:keep-alive<cr><lf><cr><lf>
```

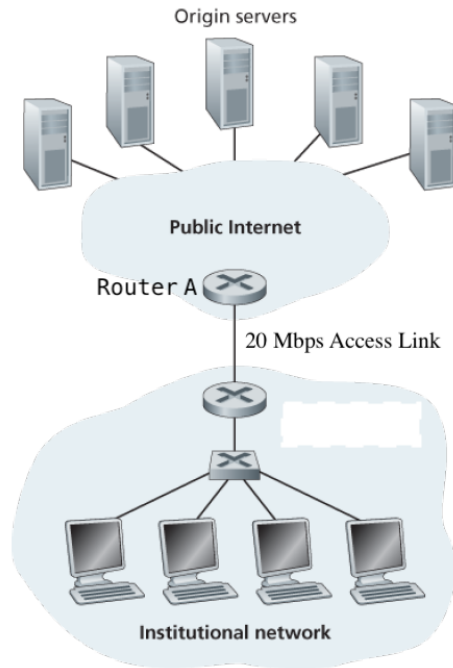
- Does HTTP message run on top of TCP or UDP? Why is TCP or UDP a better choice? Please explain the reason by considering the features of TCP or UDP.
- Is this message an HTTP request message or an HTTP response message?
- Does this message corresponds to a non-persistent or a persistent connection?
- Suppose the browser who sent this message was assigned an identification number 1150 by the associated host. What is the corresponding entry in the cookie file of the browser? To specific its cookie, which header line (including header field name and value) should be included in the message?
- If the server receives this message successfully and is going to return the requested object in a message, what would be the status line? What would be included in the entity body?

Solution:

- TCP; reliable data transfer
- HTTP request
- persistent
- gaia.cs.umass.edu 1150; cookie: 1150
- HTTP/1.1 200 OK; The requested html file at /cs453/index.html

Q 7. Consider the following figure with an institutional network connected to the Internet. There are a set of objects with size 1 Mbits. Suppose the institution's browsers has an average request rate of 10 requests per second, and all those requests are sent to the origin servers. The average response time is determined as follows:

$$\text{Average Response Time} = \text{Internet Delay} + \text{Average Access Delay}. \quad (2)$$



- Internet delay is the round trip time between router A and the origin server. It is equal to 2 seconds.
- Average access delay is the delay from Router A to the institution router. The transmission rate of the access link is 20 Mbps. The average access delay is equal to $\Delta/(1 - \Delta\beta)$, where Δ is the average time required to transmit an object over the access link (i.e., the transmission delay of an object); β is the arrival rate (in requests per second) at the access link.
- Note: The delays over other links, e.g., local area network (LAN) delay, are regarded as zero.

Answer the following questions:

- Derive average response time of the system.
- Suppose there is a cache installed in the institutional LAN. Compute the hit rate $x \in [0, 1]$ that leads to an average response time that is less than 1 second.

Solution:

- The average access delay is equal to $\Delta/(1 - \Delta\beta)$, where $\Delta = 1\text{Mbits}/20\text{Mbps} = 0.05$ second, and $\beta = 10$ requests per second. Thus, the average access delay is equal to $0.05/(1 - 0.05 \times 10) = 0.1$ second. The average response time = $0.1 + 2 = 2.1$ seconds.
- Suppose the hit rate is $x \in [0, 1]$. Then, $10x$ requests per second are served by the cache within LAN, which has a delay of zero. Meanwhile, $10(1 - x)$ requests per second are served by the origin server. Their associated response delay is equal to $0.05/(1 - 0.05 \times 10 \times (1 - x)) + 2$. Thus, the average response delay of all requests is

$$(1 - x) \times (0.05/(1 - 0.05 \times 10 \times (1 - x)) + 2) + x \times 0.$$

To ensure this value to be less than 1 second, we have $x > 0.515965$.

Q 8. Suppose you click a web page within your Web browser, and your local DNS server has stored the related resource records. Let RTT_0 denote the round trip time (RTT) between your host and your local DNS server. On the web page you visit, there are an HTTP basic file and 12 referenced objects. Let RTT_1 denote the RTT between the local host and the Web server. The HTTP basic file has a size of L Mbit, and each referenced object has a size of L Mbit. The transmission rate is R Mbps. Compute how much time elapses from when you click the link until your web browser receives the objects.

- Non-persistent HTTP with no parallel TCP connections?
- Non-persistent HTTP with the browser configured for 4 parallel connections?

- (c) Persistent HTTP? Note that in this case, if a client knows the URLs of its requested referenced objects, the client can send requests of referenced objects back-to-back without waiting for the responses.

Solution:

- (a) $RTT_0 + (2RTT_1 + L/R) + 12 \times (2RTT_1 + L/R)$
- (b) $RTT_0 + (2RTT_1 + L/R) + 3 \times (2RTT_1 + L/R)$
- (c) $RTT_0 + RTT_1 + (RTT_1 + L/R) + RTT_1 + 12 \times L/R$

Q 9. Consider distributing a file of F bits to N peers using a client-server architecture. Assume a fluid model where the server can simultaneously transmit to multiple peers, transmitting to each peer at different rates, as long as the combined rate does not exceed u_s .

- (a) Suppose that $u_s/N \leq d_{min}$. Specify a distribution scheme that has a distribution time of NF/u_s .
- (b) Suppose that $u_s/N \geq d_{min}$. Specify a distribution scheme that has a distribution time of F/d_{min} .
- (c) Conclude that the minimum distribution time is in general given by $\max\{NF/u_s, F/d_{min}\}$.

Solution:

- (a) The bandwidth is evenly distributed. That is, the server transmits to N user at the same time, and the rate of each user is u_s/N . Thus, distribution time is equal to NF/u_s .
- (b) The server transmits to N users at the same time. The rate assigned to each user is d_{min} . The distribution time is F/d_{min} .
- (c) The total bits that the server needs to distribute is NF , and the server bandwidth is u_s , therefore $t \geq NF/u_s$. The client with the lowest download rate needs time $t > F/d_{min}$. Therefore, the minimum distribution time is in general given by $\max\{NF/u_s, F/d_{min}\}$.

Q 10. Explain why a hierarchical DNS system (instead of a huge centralized server) is used.

Solution: A hierarchical DNS system is used instead of a single centralized server because a centralized design would create a single point of failure, overwhelming traffic, and slow access to a distant, massive database. The hierarchical structure spreads the load across many servers, improving scalability, reliability, and performance.